Do Now · retrieval

SQL SELECT, JOIN and nested SELECT

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the purpose of an INNER JOIN?

Q2

Which relational key connects two tables?

Q3

What does WHERE do?

Q4

Binary search for 9 in sorted [1,3,5,7,9,11,13]. Show midpoint indexes.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — SELECT names output columns, FROM identifies input tables and WHERE restricts rows using a Boolean predicate.

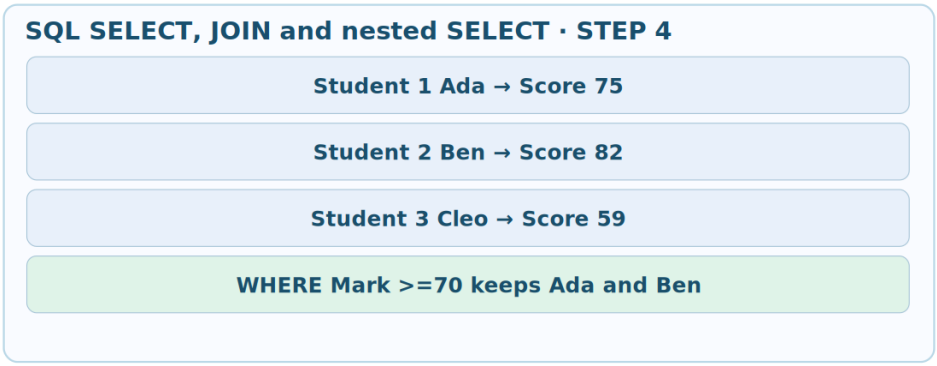
Core idea 2

Trace — INNER JOIN combines rows matching a join condition. A nested SELECT can return a set of keys used by an outer IN condition.

Core idea 3

Explain — Both example queries return Ada and Ben, but joining is more flexible when fields from both tables are required.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

SELECT — Choose fields for output.

INNER JOIN — Combine matching rows.

nested SELECT — Query inside another query.

IN — Compare value against returned set.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

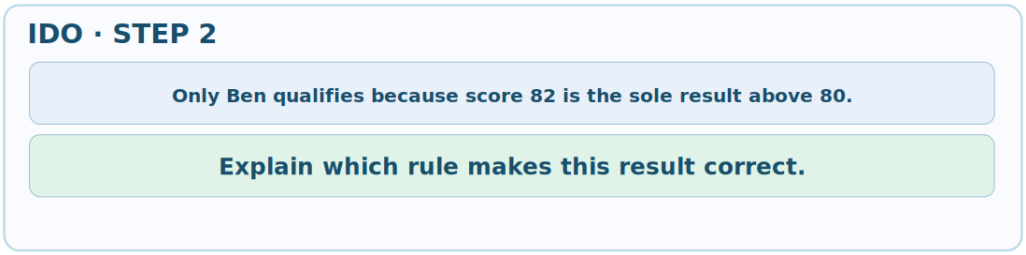
Problem

Write SQL for student names where Score.Mark>80.

Follow the worked method

1. Predict — Only Ben qualifies because score 82 is the sole result above 80.
2. Check the answer — Explain which rule makes this result correct.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict

Subquery returns IDs 1 and 2

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Write SELECT to show all Name values in Student sorted by Name ascending. [4]

Question 2

Write and trace a join returning Name and Mark for scores >=70. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Write a nested SELECT for students scoring above 75. Explain result. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

INNER JOIN results can be the same as a nested key SELECT for a particular data set.

Use key vocabulary from memory

1. Explain SELECT in your own words.

2. Explain INNER JOIN in your own words.

3. Explain nested SELECT in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.